

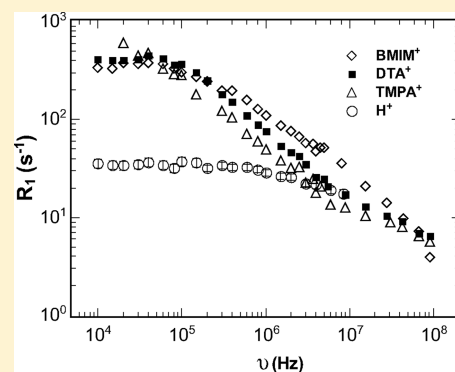
Proton NMR Relaxometry Study of Nafion Membranes Modified with Ionic Liquid Cations

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ABSTRACT: Proton nuclear magnetic resonance (NMR) relaxometry was used to study the ionic mobility and levels of confinement within Nafion membranes modified by incorporation of selected ionic liquid (IL) cations. These studies were performed aiming at understanding the effect of using different types of ionic liquid cations, and their degree of incorporation, in the values of the spin–lattice relaxation times (T_1) obtained at different values of frequency and thus detect the influence of confinement level on the ions mobility. The frequency dependence of the proton spin–lattice relaxation rate, $R_1 = 1/T_1$, for the modified Nafion/IL cation membranes was compared with that obtained for an unmodified Nafion membrane, allowing for distinguishing different contributions of the motions of the molecules depending on the frequency tested. The experimental R_1 results were analyzed in terms of models that consider the sum of the most effective relaxation contributions, to estimate the translational self-diffusion coefficient of the moving molecular species in the modified membranes. The stability of these membranes with temperature in terms of the spin–lattice relaxation was compared with results obtained by thermogravimetric analysis.



INTRODUCTION

Nafion is a polymer that combines the extremely high hydrophobicity of the polytetrafluoroethylene backbone with the high hydrophilicity of the sulfonic acid side groups arranged in intervals along the chain.¹ These acid groups, which are known to aggregate into clusters, allow for the transport of ions and thus serve as a polymer electrolyte. The chemical structure of Nafion is shown in Figure 1.

Nafion membranes have been widely used as a reference material in polymer electrolyte membrane fuel cells (PEMFCs), due to their high ionic conductivity and mechanical, thermal, and chemical stability at temperatures up to 80 °C. Additionally, due to the fact that these materials present unique properties, several studies aiming at describing their molecular structure and organization have been published in the literature, during the last 30 years.² Although several models have been proposed, they all agree that the ionic groups are aggregated in the perfluorinated polymer matrix to form a network of clusters that allow for the transport of ions and swelling by polar solvents. However, there is still no generalized agreement regarding the size, geometry, and the spatial distribution of these clusters.² These ionic clusters may be connected to each other by narrow channels, leading to the formation of different environments inside the membrane, with different levels of confinement, and can have different dimensions (between 1 and 10 nm) depending on the water content of the membrane. The level of confinement of the ionic clusters has an important effect on the degree of water structuring within the membrane matrix, which is expected to directly

influence the mechanisms of proton transport through these membranes.

While seeking to understand the structure of Nafion, there are still some problems in the performance of these materials that need to be investigated when used in PEMFCs. One of the main problems observed in Nafion membranes is the decrease in ionic conductivity at temperatures above 80 °C, which may be attributed to the loss of water contained inside the membrane, which negatively impacts on the proton mobility. To overcome the problem of water loss observed at high temperatures, a possible solution has been already proposed in a previous work³ that consists of the modification of Nafion membranes by incorporation of ionic liquid (IL) cations. Ionic liquids (ILs) are compounds consisting entirely of ionic species comprising an organic cation and either an inorganic or organic anion.⁴ They present good electrical conductivity, high ionic mobility, and good thermal and chemical stability.^{5,6} In the aforementioned work,³ Nafion membranes were modified by partially replacing the H^+ of the membrane by an IL cation, thus leading to different concentrations of IL cation incorporated inside the membrane. It has been observed that the ionic conductivity of the modified Nafion/IL cation membranes may be tuned according to the type of IL cation incorporated as well as their incorporation degree inside the membrane. The most important result obtained with

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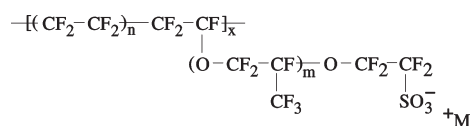


Figure 1. Chemical structure of Nafion.

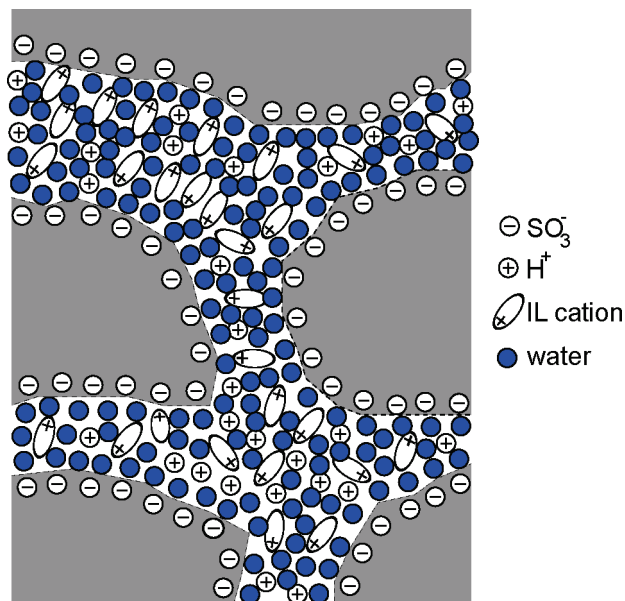


Figure 2. Schematic representation of the Nafion/IL cation membrane.

these modified membranes was the higher stability observed at temperatures up to 200 °C. The modified membranes were also characterized in terms of distribution of the IL cation within the superficial layer of the membrane using X-ray photoelectron spectroscopy (XPS); their electric properties were determined by electrochemical impedance spectroscopy (EIS), and stability at high temperatures was evaluated by thermogravimetric analysis. However, the effect of incorporating IL cations into Nafion on the global ion mobility and its impact in the resulting levels of confinement within the membrane are not yet fully understood. A possible schematic representation of a Nafion membrane incorporated with the IL cations is shown in Figure 2, where the protons and the ILs cations are distributed inside clusters with different dimensions, as well as water molecules.

In this work, nuclear magnetic resonance (NMR) relaxometry measurements were performed for Nafion and for modified Nafion/IL cation membranes to study the mobility of the cations and infer on their levels of confinement. This experimental technique has been successfully used for the study of molecular dynamics in liquid crystals in confined environments^{7–10} and also for characterization of proton mobility in Nafion membranes with different degrees of water content.^{11–13}

The proton NMR relaxation technique describes the energy exchange between the proton spins of the hydrogen nuclei and the energy exchange between this spin system and the surrounding lattice. These energy exchanges are characterized by two relaxation rates: the spin–spin relaxation rate, $R_2 = 1/T_2$, and the spin–lattice relaxation rate, $R_1 = 1/T_1$, where T_2 is defined as the spin–spin relaxation time and T_1 as the spin–lattice relaxation time, respectively. In this work only the spin–lattice relaxation time (T_1) will be analyzed. When T_1 is measured on a broad

range of Larmor frequencies (ν), it is possible to obtain detailed information regarding the molecular dynamics of the system.⁸ The Larmor frequency is defined as $\nu = \gamma B/2\pi$, where B is the magnetic field and γ is the proton gyromagnetic ratio. It should be mentioned that spin–lattice relaxation measurements in the rotating frame, $T_{1\rho}$, can also provide information about slow molecular motions.¹⁴ However, the frequency range accessible by this NMR technique is considerably smaller than the one accessible by fast field-cycling NMR.

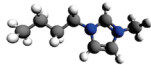
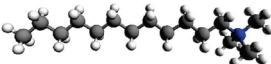
The use of ^1H NMR relaxometry to study the ionic liquid cation motions inside Nafion membranes presents a clear advantage with respect to other NMR spectroscopy studies using other nuclei (e.g., ^{13}C , ^{19}F) because only the molecules that can diffuse possess hydrogen in their chemical structure. Therefore, the NMR signal reflects only the motion that involves hydrogen proton spins. In the case of water-soluble ionic liquids, the motion of water molecules on the NMR signal can be additionally partially suppressed if deuterated water is used to prepare the ionic liquid solutions, as the resonance frequency of ^2H is different from the proton Larmor frequency. However, the water signal suppression is never complete as the Nafion membrane in the protonated form already contains water in its structure that cannot be totally removed even after the incorporation of the IL cations. In any case, according to the schematic representation of the spatial distribution of the moving molecular species in the Nafion/IL cation membrane shown in Figure 2, we have to consider the motions inside a layer close to the cluster's surfaces defined by the sulfonic groups of the polymeric matrix and the “almost” bulklike motions within the clusters inner volume.

This experimental technique is directly sensitive to translational self-diffusion of molecules because the ^1H spin–lattice relaxation depends on dipolar spin interactions that can occur between *inter-* or *intramolecular* spins.

The use of pulse field gradient (PFG) spin echo techniques to measure the molecular translational self-diffusion coefficients is limited to molecular displacements between 100 Å and 10 μm .¹⁵ Recent self-diffusion studies were performed in ionic liquids in Nafion membranes, using PFG NMR diffusometry, performed with diffusion times in the range 30–600 ms and root-mean-square displacements in the range 0.5–3.3 μm .¹⁶ In the particular case of the ionic liquid $\text{BMIM}^+\text{BF}_4^-$ modified membranes, the apparent self-diffusion coefficients obtained do not evidence any type of restricted-diffusion processes related with the heterogeneous nature of the Nafion/IL modified membranes. In view of the estimated size of the ionic clusters and interconnecting channels (<100 Å), the effect of medium heterogeneity on the diffusion processes might only be detected by performing a detailed study of the molecular dynamics of the moving species in the Nafion/IL membranes.

The proton NMR relaxometry results obtained for a wide range of frequencies will be used to characterize the modified Nafion/IL cation membranes in terms of the ion mobility and levels of confinement, in comparison with the results obtained for an unmodified Nafion membrane. The translational self-diffusion coefficient of IL cations are estimated by fitting relaxation models to the experimental R_1 dispersion. Fast rotations/reorientations of the molecules, translational self-diffusion, and rotations mediated by translational displacements are motions usually considered with success on the interpretation of the proton spin–lattice relaxation dispersion, obtained for low molecular mass liquid crystals confined in controlled nano porous glasses.¹⁰ Finally, the effect of increasing temperature on the stability of the

Table 1. Characteristics of the Ionic Liquid Cations Studied

Ionic liquid cation	Molecular formula	Molecular structure	Molecular weight (g/mol)	Molecular length, (Å ^a)
Phenyltrimethylammonium, TPA ⁺	(CH ₃) ₃ NC ₆ H ₅ ⁺		136	7.2
1- <i>n</i> -Butyl-3-methylimidazolium, BMIM ⁺	C ₈ H ₁₅ N ₂ ⁺		139	10.2
<i>n</i> -Dodecyltrimethylammonium, DTA ⁺	C ₁₂ H ₂₅ N(CH ₃) ₃ ⁺		228	18.2

^a Estimated by using the Avogadro software (<http://avogadro.openmolecules.net>).

modified Nafion/IL cation membranes was also assessed through NMR relaxometry, making possible a qualitative correlation of the transient behavior of the relaxation rate with water content, and their comparison with the results obtained by thermogravimetry.

MATERIALS AND METHODS

Materials. Different ionic liquid (IL) cations were incorporated in a Nafion-112 membrane, starting from its initially protonated form (H⁺) (obtained from Dupont). The Nafion-112 membrane in the protonated form has a maximum ion exchange capacity (IEC) of 0.92 mequiv/g (i.e., 0.92 mmol of H⁺ per gram of polymer) (data provided by Dupont).

The pure compounds used as source materials for the IL cations are listed below:

- phenyltrimethylammonium chloride, abbreviated as TPA⁺Cl[−], from Fluka
- 1-*n*-butyl-3-methylimidazolium bis(trifluoromethanesulfonimide), known as BMIM⁺BF₄[−] from Solchemar
- *n*-dodecyltrimethylammonium chloride, DTA⁺Cl[−] for short, from TCI Instruments

The molecular formula, molecular structure, molecular weight, and molecular length of the corresponding IL cations studied are listed in Table 1. These IL cations were selected to study the effect of the cations molecular structure and water solubility on the incorporation and degree of confinement in the membranes because DTA⁺ is a water-soluble cation, BMIM⁺ is an ionic liquid whose DTA⁺ concentration is very low, and TPA⁺ is an ionic liquid cation that contains an aromatic ring in its structure. It should be mentioned that TPA⁺ might be the most interesting cation to be incorporated in Nafion, as other studies evidence that it contributes to an increase in the ionic conductivity and thermal stability, at high temperatures (up to 200 °C) of the Nafion/TPA⁺ membrane in comparison with the unmodified Nafion/H⁺.³

Incorporation of Ionic Liquid (IL) Cations in Nafion Membranes. To remove impurities from the Nafion membrane in the protonated form, the following procedure was carried out. First, the Nafion membrane was immersed in H₂O₂ aqueous solution 3% (v/v) during 1 h, at 80 °C, followed by 1 h at the same temperature in deionized water. Then, the membrane was immersed in a solution of H₂SO₄ 1 M during 1 h at 80 °C, and finally in deionized water during 1 h at the same temperature.

After the Nafion membrane pretreatment, the ionic liquid cations were incorporated into Nafion by immersing the membranes in a solution that contains the IL cation, during different

time periods (between 1 and 40 h). Different degrees of cations concentration in the membrane can be obtained, depending on the contact time between Nafion and the IL solution, due to a cation exchange process between the H⁺ protons in the membrane and the IL cations. The incorporation degree of the IL cation (i.e., the percentage of H⁺ in Nafion that was exchanged by the IL cation with respect to the maximum ion exchange capacity) was determined by measuring the pH and the conductivity of the liquid phase, with a pH conductivity meter (model 720 A, Orion) and with a laboratory conductivity meter (model 960, Schott Instruments), respectively.

To incorporate the *n*-dodecyltrimethylammonium (DTA⁺) and the phenyltrimethylammonium (TPA⁺) cations into Nafion, it was necessary to prepare a 40% (w/w) IL aqueous solution, because the starting chemical products DTA⁺Cl[−] and TPA⁺Cl[−] are solid at room temperature. Two solutions of DTA⁺Cl[−] were prepared, one with deionized water and the other with deionized deuterated water.

Regarding the incorporation of the DTA⁺ cation, two different incorporation degrees were obtained, 13% and 68% after a 2 and 40 h of contact time between the Nafion membrane and the IL solution, respectively. For the BMIM⁺ and TPA⁺ cations, after 24 h of incorporation time, a 18% and 95% incorporation degree was achieved, respectively.

The estimated ratios between the number of water and sulfonic molecules, λ , in the modified membranes were: 12.4 for Nafion/DTA⁺, 11.9 for Nafion/BMIM⁺, and 0.4 for Nafion/TPA⁺.

NMR Relaxometry Experiments. The proton spin–lattice relaxation time (T_1) was measured as a function of the Larmor frequency between 10 kHz and 90 MHz at a temperature of 23 °C for the unmodified Nafion membrane and for the modified Nafion/IL cation membranes. In addition, the T_1 dispersion of the pure BMIM⁺BF₄[−] was also obtained, taking advantage of the fact that BMIM⁺BF₄[−] presents itself in a liquid phase.

Two different spectrometers were used. T_1 measurements at Larmor frequencies between 10 kHz and 8 MHz were performed in a home-built fast field-cycling spectrometer,^{17–19} with a polarization and detection field of 0.21 T (corresponding to the proton Larmor frequency of 8.9 MHz) and switching times of 2–3 ms. The T_1 data between 8 and 90 MHz were obtained in a conventional pulsed NMR Bruker spectrometer with a 0–2.1 T electromagnet and an Avance II 300 console, using the inversion–recovery radio frequency pulse sequence with phase cycling $((\pi)_x - (\pi/2)_{x-y})$ for suppression of the dc bias. It is important to mention that at 8 MHz, the common frequency value of the

two techniques, a good agreement between the T_1 values measured by the conventional and by the fast field-cycling spectrometer was achieved. The experimental error in the spin–lattice relaxation time measurements is estimated to be lower than $\pm 10\%$.

In addition, for one specific value of Larmor frequency (1500 kHz) proton spin–lattice relaxation (T_1) measurements were performed for the temperatures: 23, 80, and 120 °C.

RESULTS AND DISCUSSION

NMR Relaxometry Results. In Figure 3 it is presented the spin–lattice relaxation rate ($R_1 = 1/T_1$) for Larmor frequency values between 10 kHz and 8 MHz, for a Nafion/ H^+ membrane in its protonated form, with a $\lambda \sim 3.3$, at 23 °C. It is important to mention that the magnetization decay behavior for this membrane, at frequencies up to 8 MHz, is well described by a monoexponential behavior characterized by a single T_1 . Above this frequency, evidence was found of a two exponential decay of the longitudinal magnetization. This behavior could be possibly associated with a decoupling of the spin–lattice relaxation between two proton spin subsystems. Further work needs to be performed to investigate this feature. In this work, only those systems characterized by a monoexponential magnetization decay were considered.

The relaxation rates observed in Figure 3 are associated with all the hydrogen nuclei present in the membrane. For Nafion in the protonated form, the relaxation rates correspond to both 1H associated with the sulfonic groups (Figure 1) and those of water enclosed in the membrane. The R_1 values range and dispersion presented in this work are consistent with those of Perrin et al.¹³ In fact the R_1 results presented here are slightly larger than those presented in the work of Perrin et al.¹³ This may be due to the possible presence of some remaining paramagnetic impurities in the membrane that were not removed during the pretreatment described in the previous section. The presence of these paramagnetic particles is usually associated with an increase of the R_1 as described in the literature for systems doped with selected paramagnetic metal complexes.²⁰ A qualitative analysis of the results shows that for frequency values up to 3 MHz, there is practically not much variation in R_1 values with an increase of frequency. Above the frequency of 3 MHz, a slight decrease in R_1 is observed. Generally, NMR experiments enable the detection of motions of molecules with large correlation times at low frequencies, while at high frequencies it is possible to detect the molecular motions with short correlation times. Therefore, the independent frequency behavior of the relaxation rate observed for Nafion in the protonated form at low frequencies may be attributed to the fact that only motions with short correlation times are present in this membrane.

Finally, the relaxation rate values obtained for this membrane, which are in the range between 10 and 35 s^{-1} , are comparatively much larger than the characteristic relaxation rate of bulk water, 0.3 s^{-1} .¹³ Such higher relaxation rates suggest that the protons are enclosed in a confined micro environment.

As referred to above, proton spin–lattice NMR measurements were performed for modified Nafion/IL cation membranes, with different degrees of cations concentration. Experiments were carried out for a Nafion membrane incorporated with the *n*-dodecyltrimethylammonium (DTA^+) cation. The evolution of the measured spin–lattice relaxation time (T_1) with the

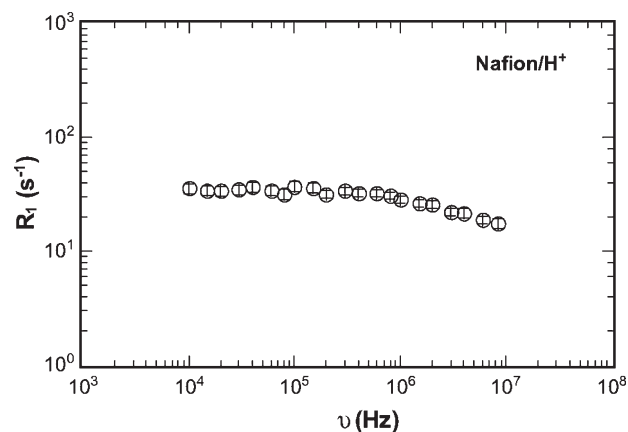


Figure 3. Frequency dependence of the proton spin–lattice relaxation rate R_1 for Nafion/ H^+ .

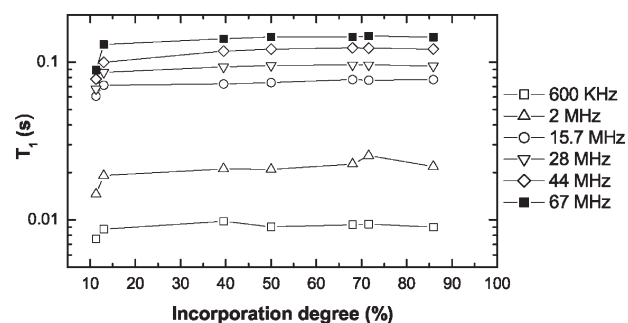


Figure 4. Spin–lattice relaxation time (T_1) as a function of the incorporation degree of DTA^+ for different values of frequency.

incorporation degree for the modified Nafion/ DTA^+ membrane, for different values of frequency is shown in Figure 4.

The magnetization decay behavior for this membrane, at frequencies up to 90 MHz, is also described by a monoexponential behavior characterized by a single T_1 . For all frequencies considered, an increase in the value of T_1 was observed with an increase in the incorporation degree, followed by an apparent stabilization of the T_1 values. It is also observed that for all values of incorporation degree, the magnitude of T_1 depends on the frequency, which is a different behavior than observed previously for an unmodified Nafion membrane, where T_1 values were almost independent from frequency.

To better understand the results obtained for the modified Nafion/IL cation membranes, two different Nafion/ DTA^+ membranes were prepared, one with a low incorporation degree (13%), obtained after a contact time of 2 h, and another with the incorporation degree obtained after reaching the equilibrium (68%), which was obtained after a contact time of 40 h. The evolution of the incorporation degree obtained with contact time can be found in a previous work.³ The proton spin–lattice relaxation time was then measured for a wide range of frequencies, between 100 kHz and 90 MHz. The frequency dependence of the proton spin–lattice relaxation rate R_1 for both Nafion/ DTA^+ (13%) and Nafion/ DTA^+ (68%) is represented in Figure 5a,b, respectively. Additionally, in the same figures there are also represented the results of the R_1 , in the frequency range 400 kHz to 68 MHz, obtained for Nafion/ DTA^+ membranes prepared with deuterated water solutions, for the same incorporation

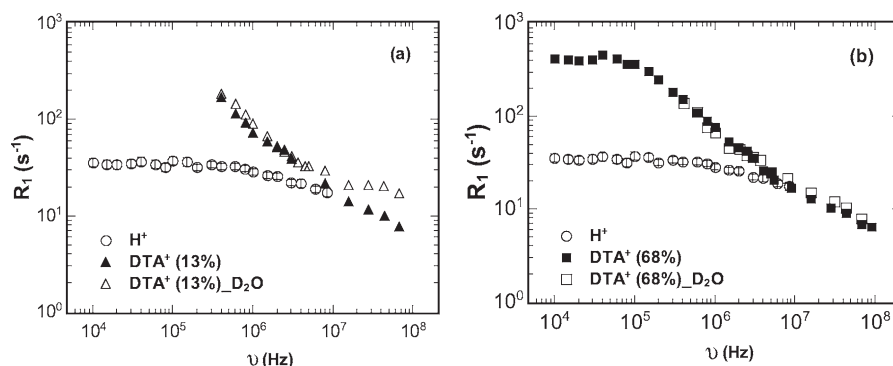


Figure 5. Comparison of the frequency dependence of the proton spin–lattice relaxation rate R_1 for (a) Nafion/H⁺, Nafion/DTA⁺ (13%), and Nafion/DTA⁺ (13%)_D₂O and (b) Nafion/H⁺, Nafion/DTA⁺ (68%), and Nafion/DTA⁺ (68%)_D₂O.

degrees of 13% and 68%, in the following referred to as Nafion/DTA⁺ (13%)_D₂O and Nafion/DTA⁺ (68%)_D₂O. The use of deuterated water makes it possible to suppress the contribution to the relaxation of the ¹H spins of the water molecules.

The relaxation rates observed in Figure 5 are associated with all the hydrogen nuclei present in the membranes. In the case of the modified Nafion/DTA⁺ membranes, in addition to the contribution of the remaining H⁺ protons, it is also necessary to take into account possible contributions of the ¹H present in the incorporated cation molecules (see Table 1 for the molecular structure of DTA⁺ cation). In both subfigures, distinct profiles may be observed for the frequency dependence of the spin–lattice relaxation rate for an unmodified Nafion membrane (Nafion/H⁺), and for that of modified Nafion/DTA⁺ membranes, prepared either with water or with deuterated water. In contrast with the Nafion/H⁺ membrane, both modified Nafion/DTA⁺ membranes show a frequency dependence of the spin–lattice relaxation rate in the full range of frequencies tested. The frequency-dependent behavior observed for Nafion/DTA⁺ membranes at lower frequencies, in contrast with the flat profile that was observed for a Nafion membrane in the protonated form, may be attributed to the existence of IL cation motions with large correlation times (slow motions).

The Nafion/DTA⁺ (13%) and Nafion/DTA⁺ (68%) present identical R_1 dispersion profiles. In the case of the modified membranes prepared with deuterated water, the R_1 dispersion is different and depends of the degree of IL cation incorporated. In Figure 5a, for a Nafion membrane with a DTA⁺ cation incorporation degree of 13%, the R_1 dispersion for frequencies below 8 MHz is similar for Nafion/DTA⁺ (13%) and for Nafion/DTA⁺ (13%)_D₂O. However, above 8 MHz this is not the case and the R_1 of Nafion/DTA⁺ (13%)_D₂O presents a leveling off that is not present for Nafion/DTA⁺ (13%) membrane. This difference seems to be directly correlated to the amount of water incorporated in the membrane. As explained previously, the use of deuterated water molecules suppresses the contribution of the water molecules spin relaxation. Because the R_1 dispersion at high frequencies is usually associated to local molecular motions, this result may indicate that in this frequency range the motion of free water molecules in Nafion/DTA⁺ (13%) membrane contributes significantly to the spin–lattice relaxation.

In the case of a Nafion membrane for which the degree of cation incorporated is at equilibrium (68%) (Figure 5b) no significant differences on the spin–lattice relaxation rates are

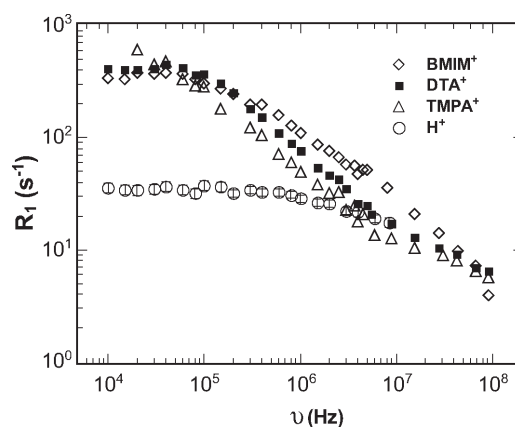


Figure 6. Comparison of the frequency dependence of the proton spin–lattice relaxation rate R_1 for the unmodified and modified Nafion/IL cation membranes.

observed for the membranes Nafion/DTA⁺ (68%) and Nafion/DTA⁺ (68%)_D₂O in the whole range of frequencies tested. This similarity is not associated with a reduced water content in the membranes in comparison with the case for the Nafion/DTA⁺ (13%) membrane. In a previous work³ it was observed that even for a Nafion membrane with the maximum incorporation degree of the DTA⁺ cation, there is still a large amount of water molecules inside its structure deeply involved in the solvation of the incorporated DTA⁺ cations. Due to this fact, it is reasonable to consider that water molecules assume an important role in the mobility and/or organization in the case of a higher DTA⁺ incorporation degree. This result may indicate that the motion of free water molecules in Nafion/DTA⁺ (68%) membrane could not be detected. If this is the case, then the R_1 dispersion for frequencies above 8 MHz reflects mainly the DTA⁺ motions affected by the confinement conditions inside the membranes.

Nafion membranes were also modified with the cations TMPA⁺ and BMIM⁺. These membranes were prepared with the maximum degree of incorporation at equilibrium, respectively, 95% and 18% for TMPA⁺ and BMIM⁺.³ A comparison of the frequency dependence of the proton spin–lattice relaxation rate R_1 is shown in Figure 6 for the unmodified Nafion membrane and for the modified Nafion membranes tested with all different cations, for the situation of corresponding maximum incorporation degree.

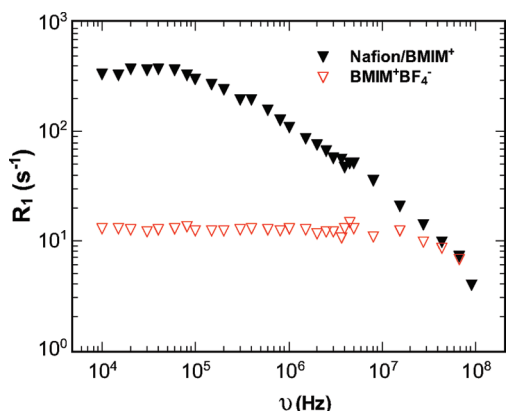


Figure 7. Comparison of the frequency dependence of the proton spin–lattice relaxation rate R_1 for the $\text{BMIM}^+\text{BF}_4^-$ and the Nafion/ BMIM^+ modified membrane systems.

A monoexponential magnetization decay characterized by a single T_1 was observed for Nafion incorporated with the TMPA^+ and BMIM^+ cations, in the whole frequency range analyzed. At high frequencies, the R_1 dispersion is similar for all the modified Nafion/IL membranes. A possible explanation for this behavior is that at high frequencies mainly the fast molecular motions affect the spin–lattice relaxation. Molecular fast rotation/reorientations and/or bulklike translational self-diffusion motions are less affected by the confinement conditions of the Nafion membranes and might be less dependent on the particular molecular structure of each IL cation.

It is interesting to note that the dispersion in the frequency range between 200 kHz and 10 MHz is clearly different for the three Nafion/IL modified membranes. At low frequencies, below 100 kHz a leveling off is observed for BMIM^+ and DTA^+ Nafion/IL systems. No leveling-off was observed in the Nafion/ TMPA^+ system within the resolution of our fast-field cycling NMR relaxometer.¹⁷

In Figure 7 the R_1 dispersion on both $\text{BMIM}^+\text{BF}_4^-$ and modified Nafion/ BMIM^+ systems is presented. It is interesting to note that at high frequencies the relaxation dispersion is quite similar in both systems. This aspect might be correlated with the fact that at these frequencies R_1 is dominated by local rotations/reorientations of the BMIM^+ cations and is less affected by the translation diffusion. In view of the different hydration states of BMIM^+ cations in both systems (very small in the $\text{BMIM}^+\text{BF}_4^-$ ionic liquid and larger in the Nafion/ BMIM^+ membrane), it is also interesting to note that the contribution of water molecules to the measured R_1 is not clearly detected in the membrane system.

Modeling of Proton Spin–Lattice Relaxation of Modified Nafion/IL Cation Membranes. The proton spin–lattice relaxation depends on the dipolar interaction between the hydrogen nuclear spins. These dipolar interactions occur between *intra*- or *intermolecular* spins but in both cases they are modulated by molecular motions. In view of the conceptual representation of the molecular species moving in the Nafion/IL cation membrane clusters, different types of motions have to be considered, taking into account the specific constraints imposed by the clusters walls defined by the sulfonic groups. The proton NMR relaxometry studies conducted in these systems detect not only motions of free protons and water molecules but also the motion of IL cations. The water dynamics of the Nafion membranes has been reported by Perrin et al.¹³ These studies show that the R_1 dispersion, in particular for frequencies below 10 MHz, is strongly dependent on the membranes water content. In fact, for hydrations

corresponding to $\lambda < 4$ the dispersion is clearly larger than for $\lambda > 4$ where the R_1 tends to a plateau at low frequencies. The motion of water molecules in other confined environments like nanotubes show also a strong spin–lattice relaxation dispersion at low frequencies.²¹ In this case the dispersion was associated with the 1D character of the water molecules inside the narrow nanotubes.

As referred to in the previous section in the case of the modified Nafion/IL cation membranes studied in this work, the R_1 dispersions observed are dominated by the relaxation mechanisms associated with the motions of the IL cations. Therefore, the relaxation model to be considered in the interpretation of the observed spin–lattice relaxation has to consider the rodlike molecular shape of the cations and the confining constraints imposed by the cluster walls.

The study of the molecular dynamics of rodlike molecules in nanosize cavities has been performed for liquid crystals in different porous media.^{9,10,22,23} The results obtained in those studies indicate that in the frequency range 10 kHz to 100 MHz the proton spin–lattice relaxation rate dispersion can be interpreted in terms of a sum of contributions from different relaxation mechanisms. The relaxation mechanisms considered for the liquid crystal molecules confined in nanosize porous are the local rotations/reorientations of the molecules parallel and perpendicular to the molecular long axis $(R_1)^R$, translational self-diffusion $(R_1)^{SD}$, and collective motions. Depending on the surface to volume ratio the molecular dynamics of rodlike molecules might have to include a contribution that results from the interaction between the molecules and the confining walls as the result of the expected increase of molecular order close to porous surfaces.

The translational self-diffusion motion of the liquid crystal molecules in bulk is detected mainly by the modulation that it produces in the intermolecular dipolar spin interactions. For the molecules close to the porous surfaces the translational displacements might be also associated with some degree of reorientation. Therefore, it is reasonable to assume the existence of an interface layer where the rotations/reorientations of the rodlike molecules, probed by intramolecular spin interactions, become dependent on the self-diffusion.

The bulklike rotations/reorientations relaxation mechanism of liquid crystal molecules usually considers two correlation times (three in more detailed studies), to take into account the movements of the molecules along the short and long molecular axis, τ_S and τ_L , respectively. Very often it is difficult to clearly identify the influence of the two correlation times. In most cases, in proton NMR relaxometry it is only possible to clearly detect the largest of these two correlations times.⁸ In any case, the relaxation model is a sum of Lorentzian curves with distribution of weights that depends, in particular, on the intramolecular inter ^1H spin distances, angles between inter ^1H spin vectors and the magnetic field, correlation times.

The spin–lattice relaxation model most often used to describe the relaxation mechanism by translational self-diffusion of rodlike molecules in the isotropic bulk phase is the one proposed by Torrey.^{8,24} This model considers the density of spins, n , a correlation time (i.e. mean jump time), $\tau_D = \langle r^2 \rangle / (6D)$, and the distance of closest approach between molecules, d , as the most relevant physical parameters. This model is usually expressed in terms of $\alpha = \langle r^2 \rangle / (12d^2)$ and in the low frequency limit $(R_1)^{SD} \sim nD^{-1}$. Typically, the mean-square jump distance $\langle r^2 \rangle \sim 1 \text{ nm}^2$ for diffusion coefficients $D = 5 \times 10^{-11} \text{ m}^2/\text{s}$ for low molecular weight rodlike liquid crystal molecules.¹⁰

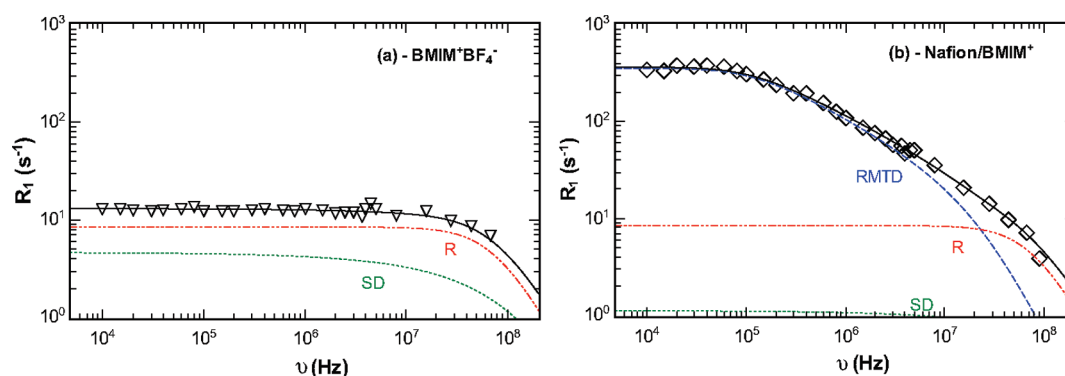


Figure 8. Experimental results and theoretical fitting curves for (a) $\text{BMIM}^+\text{BF}_4^-$ and (b) $\text{Nafion}/\text{BMIM}^+$.

In the present case the contribution of rotations/reorientations will be approximated by

$$(R_1)^R = (R_1)_0^R \frac{1}{5} \left[\frac{1}{1 + \omega^2 \tau_c^2} + \frac{4}{1 + 4\omega^2 \tau_c^2} \right] \quad (1)$$

where $(R_1)_0^R$ is the low frequency asymptotic value and τ_c is an effective correlation time. In the high frequency region where $\omega \tau_c \gg 1$, $(R_1)^R \sim (R_1)_0^R / (\omega^2 \tau_c^2)$.

In the case of small rodlike molecules confined in nanoscale cavities, the molecular motions close to the confining walls are restricted and the rotations/reorientations and the translational displacements are affected mutually. This type of restricted motion is generally referred to as rotations mediated by translational displacements (RMTD). In fact, it is reasonable to assume that the IL cation molecules might interact with the sulfonic groups of the Nafion membrane. This interaction might induce some alignment of the IL cation molecules because the cation's polar heads will be close to the negative charges of the sulfonic groups and the cation's tails will be away from the clusters/channels surfaces. Therefore, an interface layer is formed where the molecular order might be slightly higher than that observed closer to the center of the clusters. The thickness of this layer, the fraction of ordered molecules, and the orientational structure factor must be taking into account when the contribution of this relaxation mechanism to the spin–lattice relaxation rate is calculated.⁹

The translational displacements close to the clusters defining surfaces can be analyzed in terms of diffusion modes with different wave numbers q , decaying in time with dumping constants $\tau_q = (Dq^2)^{-1}$, each expressed by a Lorentzian-type of curve. The relaxation rate for this mechanism is the sum of all diffusion modes taking into account that each mode might contribute with a different weight. The weights depend on the structure of the alignment field in the porous matrix. The orientational structure factor is, in general, a complex function of the wavenumber, but in same cases it can be approximated by a power law with an exponent p . If all modes are equally weighted, then $p = 1/2$. The spin–lattice relaxation rate associated with this mechanism for an arbitrary value of p was calculated for a liquid crystal of rodlike molecules confined in glasses with ~ 15 nm porous:

$$\begin{aligned} \left(\frac{1}{T_1} \right)^{\text{RMTD}} &= (R_1)^{\text{RMTD}} \\ &= A_{\text{RMTD}} \left\{ \frac{1}{\omega^p} \int_{z_{\min}}^{z_{\max}} \frac{z^{3-2p}}{1+z^4} dz + \frac{1}{(2\omega)^p} \int_{z_{\min}/\sqrt{2}}^{z_{\max}/\sqrt{2}} \frac{z^{3-2p}}{1+z^4} dz \right\} \end{aligned} \quad (2)$$

where $z_{\max} = (\omega_{c\max}/\omega)^{1/2}$ and $z_{\min} = (\omega_{c\min}/\omega)^{1/2}$. A_{RMTD} is a parameter that depends on the residual dipole–dipole proton spin interaction averaged by the local molecular reorientations, on the microstructural features of the confined environment, and on the diffusion coefficient.⁹ The high and low cutoff frequencies, $\omega_{c\max}$ and $\omega_{c\min}$, respectively, are related to the self-diffusion constant (D) through the following relations: $\omega_{c\max}^{-1} = l_{\min}^2/4D$ and $\omega_{c\min}^{-1} = l_{\max}^2/4D$, where $l_{\max}$ and $l_{\min}$ are respectively the largest and smallest displacement distances.¹⁰

As previously mentioned, modified Nafion/IL cation membranes appear to present nanosized cavities interconnected by narrow channels where the small species (e.g., H^+ , IL cations, and H_2O molecules) move. Therefore, the different dimensions of cavities and channels inside the membrane may influence the proton mobility in the membrane, because there are regions with different confinement conditions. Additionally, the presence of water molecules and their degree of structuring, which results from the incorporation of IL cations, may also influence the ion mobility inside the membrane. The motion of the molecules inside Nafion can therefore be considered similar to that of liquid crystal molecules confined in nanoporous glasses.¹⁰ The approach followed in this work involved the fitting of a model equation, given by a linear combination of (R + SD) and RMTD models, to the experimental spin–lattice relaxation rate data. A similar model was used to analyze the proton spin–lattice relaxation dispersion obtained for confined liquid crystals in nanoporous glasses,^{9,10} where the frequency dependence of the spin–lattice relaxation rate is similar to that observed in this work (Figures 5 and 6). The fitting model equation is thus given by

$$R_1 = [(R_1)^R + (R_1)^{\text{SD}}] + (R_1)^{\text{RMTD}} \quad (3)$$

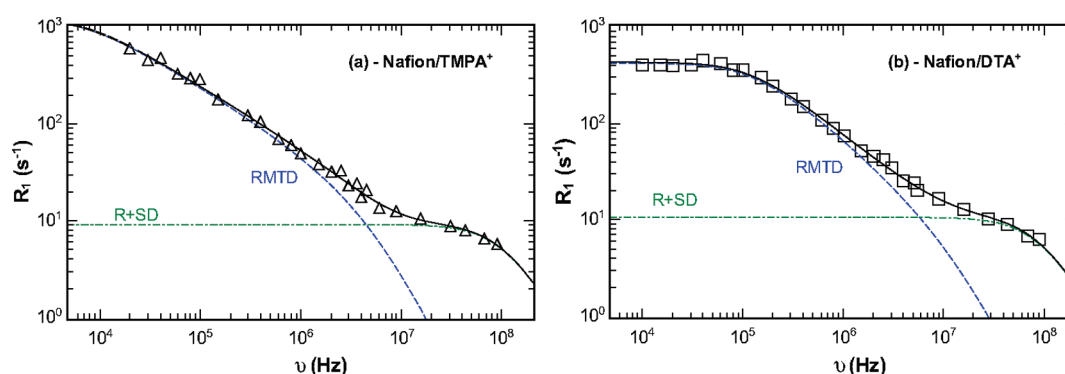
Because all ^1H spins in the molecules contribute to the relaxation without distinction, the three contributions to the total relaxation rate have equal weights.

Equation 3 was fitted first to the experimental results of both $\text{BMIM}^+\text{BF}_4^-$ and $\text{Nafion}/\text{BMIM}^+$ simultaneously.²⁵ This fit was performed by assuming that local rotations/reorientations in both systems are described by the same correlation time τ_c and the prefactor $(R_1)_0^R$ in eq 1. It was further assumed that in $\text{BMIM}^+\text{BF}_4^-$ the self-diffusion contribution was calculated with the values $d = 3.5$ Å and $n \sim 4 \times 10^{28}$ spins m^{-3} and by assuming $D^{\text{iso}} \sim 2 \times 10^{-11}$ $\text{m}^2 \text{s}^{-1}$ from the literature.²⁶ The RMTD contribution was considered only in the case of the $\text{Nafion}/\text{BMIM}^+$ systems. The best fits are shown in Figure 8 and the fitting parameters are presented in Table 2. p , D , $\omega_{c\min}$, $\omega_{c\max}$, $(R_1)_0^R$, and τ_c were considered free fitting parameters. The contributions of SD and R considered in the

Table 2. Parameter Values Obtained for the Best Fits of the Spin–Lattice Relaxation Model to the Experimental Results As Explained in the Text^a

	BMIM ⁺ BF ₄ [−]	Nafion/BMIM ⁺	Nafion/TMPA ⁺	Nafion/DTA ⁺
<i>p</i>		0.50 ± 0.06	0.62 ± 0.04	0.92 ± 0.05
<i>A</i> _{RMTD} (10 ⁴ s ^{−(1+p)})		3.0 ± 0.9	10 ± 5	1062 ± 764
$\omega_{c\min}/(2\pi)$ (10 ³ Hz)		159 ± 17	16 ± 9	161 ± 6
$\omega_{c\max}/(2\pi)$ (10 ⁶ Hz)		37 ± 5	11 ± 3	25 ± 9
τ_c (10 ^{−9} s)	1.18 ± 0.07	1.18 ± 0.07	0.8 ± 0.2	0.9 ± 0.4
(<i>R</i> ₁) ₀ ^{R+SD} (s ^{−1})	8.4 ± 0.2	8.4 ± 0.2	9.1 ± 0.9	10.6 ± 0.4
<i>l</i> _{min} (10 ^{−9} m)		1.02	0.7	1.8
<i>l</i> _{max} (10 ^{−9} m)		18 ± 2	19 ± 7	22 ± 4
<i>D</i> (10 ^{−11} m ² /s)		8.3 ± 0.9	0.9 ± 0.3	13 ± 5
<i>D</i> ^{iso} (10 ^{−11} m ² /s)	2 ²⁶			

^a The fitting and the calculated parameters are presented in the upper and bottom parts of the table, respectively.

**Figure 9.** Experimental results and theoretical fitting curves for (a) Nafion/TMPA⁺ and (b) Nafion/DTA⁺.

relaxation model fitted to the experimental results explain well the observed dispersion in the pure BMIM⁺BF₄[−] system by taking into account the simplified model used the rotations/reorientations with just one correlation time.

It is interesting to see that the same *R* contribution is enough to explain the high frequency range *R*₁ dispersion of the Nafion/BMIM⁺ system. Due to the low value of the self-diffusion coefficient obtained from the RMTD contribution of the model fit to the Nafion/BMIM⁺ *R*₁ data, the contribution of a self-diffusion bulk-type contribution is very small. In fact, because the size of the membrane's channels and clusters is small (2–10 nm), this might be expected as the surface effects on the translational self-diffusion might extend to the whole clusters/channels volumes.

In view of the membrane's water content, it is reasonable to expect that the diffusion of water molecules is larger than that of BMIM⁺ cations; therefore, the (*R*₁)^{water} contribution will be small when compared with the one of BMIM⁺, (*R*₁)^{BMIM+}.^{16,27}

In the model fits to the experimental data of Nafion/DTA⁺ and Nafion/TMPA⁺ systems, the contributions of *R* and *SD* are not presented separately but are represented by a single curve (*R*₁)^{R+SD} = (*R*₁)^R + (*R*₁)^{SD}, with a prefactor (*R*₁)₀^{R+SD} and a single correlation time τ_c . In Figure 9a,b are presented the best fits obtained with eq 3 to the *R*₁ experimental results obtained for the Nafion/DTA⁺ and Nafion/TMPA⁺ systems. The symbols represent the experimental values and the solid lines represent the total relaxation rate, while the dash and dash-dot lines represent the individual contributions of the RMTD and the (*R* + *SD*) relaxation mechanisms, respectively. The values of the fitting parameters are presented in Table 2.

The fitted values of (*R*₁)₀^{R+SD} presented in Table 2 are similar for the three Nafion/IL cation membranes systems. The effective correlation time τ_c is close to 10^{−9} s for the three systems. These two aspects show that the local rotations/reorientations are similar in the three systems.

The high cutoff frequencies obtained from the model fits are different for the three membrane systems and present similar error estimates. The low cutoff frequencies could also be obtained from the model fit, but in the case of the Nafion/TMPA⁺ modified membrane this value presents a larger error estimated because no *R*₁ plateau is clearly observed at low frequencies.

Because the fitted cutoff frequency $\omega_{c\max}$ of the RMTD model is related to the translational molecular self-diffusion constant, $\omega_{c\max}^{-1} = l_{\min}^2/4D$, a global diffusion coefficient for all the molecules containing ¹H can be estimated by assuming physically reasonable values for the *l*_{min} (e.g., the molecular length of the IL cations). In addition, from the low cutoff frequency it is possible to estimate the largest displacement distance because $\omega_{c\min}^{-1} = l_{\max}^2/4D$. The values of *D*, *l*_{min}, and *l*_{max} are presented in Table 2.

Because there are different proton spins contributing to the relaxation rate (free H⁺ and ¹H of water and of IL cations), one can expect three values of $\omega_{c\max}$, one for each moving molecular species. However, with our experimental results it is only possible to detect the largest of these possible high cutoff frequencies. In view of the molecular lengths and expected diffusion constants of the free H⁺ and ¹H of water and of IL cations it is reasonable to assume that the high cutoff frequency in Table 2 corresponds only to the molecular species with largest *l*_{min} and smallest

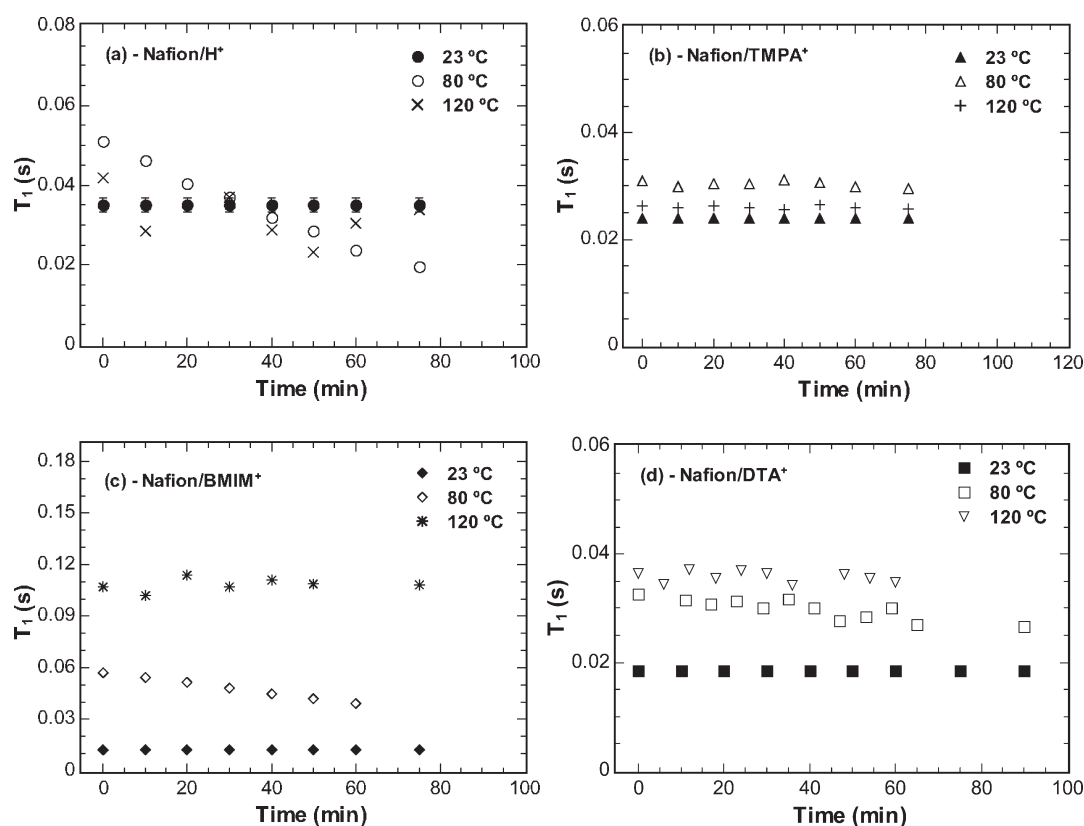


Figure 10. Time evolution of the spin–lattice relaxation time for different temperatures and a constant Larmor frequency of 1500 kHz: 23, 80, and 120 °C for (a) Nafion/H⁺, (b) Nafion/TMPA⁺, (c) Nafion/BMIM⁺, and (d) Nafion/DTA⁺ membrane.

diffusion coefficient, which is the IL cation. As mentioned before, this assumption is supported by the fact the R_1 dispersion of the modified Nafion/DTA⁺ (68%) D₂O is similar to the one observed for the Nafion/DTA⁺ membrane prepared with H₂O.

It is interesting to note that the largest displacement distances obtained from the RMTD relaxation contribution are about 20 nm, therefore larger, by a factor ~ 2 , than the reported typical size of the fully hydrated membrane cluster/channels. Similar traveling distances, $l_{\max}$, were estimated for a liquid crystal confined in 15 nm porous glasses.¹⁰ These diffusion distances are considerably smaller than those accessible by PFG NMR techniques, where the root-mean-square displacements are usually larger than 0.2 μm .¹⁶ It is therefore important to note that the apparent diffusion coefficients measured in Nafion membrane systems by PFG NMR techniques may be smaller than the ones obtained in this work because the diffusion path over distances larger than the size of a few clusters and channels must be more restricted.

The diffusion constant obtained for the Nafion/BMIM⁺ modified membrane is larger than the diffusion constant measured for the BMIM⁺BF₄[−] ionic liquid. This can be understood in view of the completely different hydration conditions of the cation molecules in the two systems. The increased apparent diffusion coefficient with an increase of hydration conditions was also reported on PFG NMR studies of Nafion/IL membranes.¹⁶

The fact that the diffusion constant estimated for the modified Nafion/TMPA⁺ membrane is the lowest when compared with those for the other modified Nafion/IL cation membranes might be related with the low water content in the membrane, as shown by the thermogravimetry results presented below.

The different values obtained for the exponent p of the RMTD relaxation mechanism (see Table 2) show that this relaxation contribution results from different distributions of translational diffusion modes and weights in the three membrane systems. Assuming that the Nafion clusters and channel walls are similar on the three systems it is clear that the effect of the Nafion walls on the translational diffusion of the IL cations depends on the molecular structure of the cations and their hydrated state. Nafion/BMIM⁺ and Nafion/DTA⁺ membrane systems present the two extreme cases with $p \sim 0.5$ and $p \sim 0.9$, respectively. The molecular length of the DTA⁺ cation and its interaction with the Nafion walls seems to disturb considerably the distribution of translational diffusion modes, possibly in correspondence with a more uniform alignment of the molecules in the surface layer along the Nafion pores and channels.

Effect of Temperature on the Spin–Lattice Relaxation Time. To study the stability of the modified membranes at high temperatures, spin–lattice relaxation time measurements were performed as a function of time for an unmodified Nafion membrane as well as for three different modified membranes with the maximum degree of incorporation, at three different temperatures, for a Larmor frequency of 1500 kHz. For each membrane the measurements were first performed at 23 °C, then at 80 °C, and, finally at 120 °C. The results obtained are shown in Figure 10.

It can be seen that, at room temperature, all membranes present a stable T_1 value throughout the experiment: this result may be taken as an indication of membrane stability in terms of their water content. For higher temperatures, the first T_1 value measured is larger when compared with the corresponding value

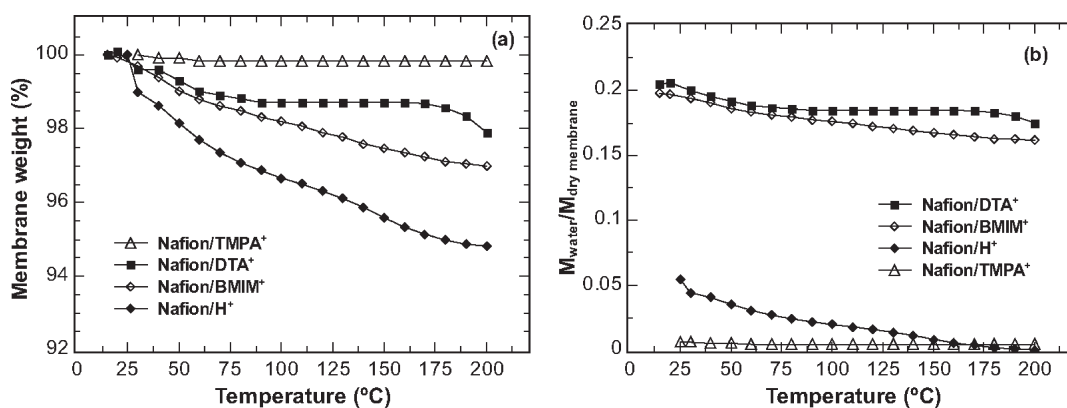


Figure 11. (a) Evolution of the membrane weight loss with increasing temperature. (b) Evolution of the relative amount of water in the membrane with increasing temperature. (Figures adapted from ref 3.)

obtained at room temperature. This behavior is expected because at high frequencies the spin–lattice relaxation time is mostly affected by bulk translational self-diffusion and fast molecular rotations/reorientations with relaxation rates, $1/T_1$, which are proportional to thermally activated correlation times.

Regarding the unmodified Nafion membrane (Figure 10a), a strong decrease of T_1 with time is observed at 80 °C, which is most possibly related with membrane dehydration at that temperature. Thermogravimetric analysis were performed for these membranes and the membrane weight loss as a function of temperature for the different modified Nafion/IL cation membranes is shown in Figure 11a, where its evolution may be related with the membrane ability to retain water and with its water degree of structuring and confinement. Therefore, the decrease observed in Figure 10a is in agreement with the thermogravimetric results present in Figure 11. For this membrane, at 80 °C is depleted all water contained within this sample, because the T_1 values at 120 °C do not present a clear trend with time and are more scattered, as a result of a low signal-to-noise ratio of the measured signals.

In the case of Nafion/TMPA⁺ (Figure 10b) a stable value of T_1 with time was obtained for all temperatures considered, which evidence the stability of this membrane even at high temperatures. This observation is also in agreement with the results obtained by thermogravimetry analysis (Figure 11a).

The results shown in Figure 10d suggest that the Nafion/DTA⁺ membrane is somewhat less stable than Nafion/TMPA⁺. In fact, a slow decrease in T_1 is observed at 80 °C, indicating dehydration of the membrane. At 120 °C the membrane appears to maintain its stability corresponding to stable T_1 values, in agreement with thermogravimetric results (Figure 11a).

The Nafion/BMIM⁺ membrane present the largest decrease of T_1 with time at 80 °C, as can be observed in Figure 10c. The measurements performed at 120 °C indicate an almost complete water depletion as far as these measurements are sensitive to.

The results obtained in this work show that the membrane stability is linked to the type and degree of cation incorporated, which ultimately determines their water content and water degree of structuring. The Nafion/TMPA⁺ membrane presents a high cation incorporation degree (95%) and a reduced water content, while Nafion/DTA⁺ exhibits a lower cation incorporation degree (68%) but a larger water content (Figure 11b). In both cases the water molecules present within the membrane seem to be highly structured, making difficult their loss at higher

temperatures, resulting in more stable membranes when compared with the case of the Nafion membrane in the protonated form. In what concerns the Nafion/BMIM⁺ membrane, it has a low cation incorporation degree (18%) and relatively high water content. The water present within this membrane appears to be less confined and structured, in comparison with the other modified membranes tested, which is confirmed by its lower stability at higher temperatures.

CONCLUSIONS

This work presents a proton NMR relaxometry characterization study of the ion mobility in Nafion membranes and in modified Nafion/IL cation membranes.

The molecular dynamics in the modified Nafion/IL cations membranes was analyzed by performing measurements of the spin–lattice relaxation times (T_1) at different values of Larmor frequency. The experimental results showed that it is not possible to detect slow motions of molecules in a Nafion membrane in its protonated form, while for modified membranes the contributions associated with both slow and fast motions of the molecules present in the membranes could be detected.

The frequency dependence of the spin–lattice relaxation rate (R_1) data obtained for the modified Nafion/IL cation membranes was interpreted in terms of relaxation model that considers the sum of rotations/translational self-diffusion and of rotations mediated by translational displacements. It was concluded that the approach followed in this work for modified Nafion membranes with IL cations results in membranes with two distinct environments: one more confined and with a lower proton mobility, described by the RMTD model (low frequencies), and another less confined, with a higher proton mobility, described by the (R + SD) model (high frequencies). By using the best fit parameters obtained with this model, it was possible to estimate diffusion coefficients for the IL cations. Different distributions of translational diffusion modes in the RMTD relaxation mechanism were detected depending on the molecular structure of the IL cations. The results obtained suggest that the Nafion membrane incorporated with the TMPA⁺ cation has a more confined and structured environment, when compared with Nafion/BMIM⁺ and Nafion/DTA⁺ membranes, because its estimated diffusion coefficient is 1 order of magnitude lower, when compared with the values obtained for the other two membranes.

The stability of the membranes was also analyzed in terms of the time evolution of the T_1 relaxation at different temperatures. The results obtained are in agreement with those obtained in previous thermogravimetric studies for these membranes that point to a strong reduction of the proton mobility at high temperatures on the unmodified membrane, while for the modified membranes, especially for Nafion/TMPA⁺ and Nafion/DTA⁺, the ion mobility presents small variations with increasing temperature.

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